

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University, New Delhi -110067
Computer Fundamental (CS-402)

Total Marks: 20

Date: 07.02.2022

Time: 2hr

All Compulsory

5. Express the following function in a) some of minterms and b) product of maxterms.

d) $F(x,y,z)=1$

e) $F(w,x,y,z)=y'z+wx y'+wxz'+w'x'z$

f) $F(x,y,z)=(xy+z)(y+xz)$

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6. For the function ; $F=xy+xy'+y'z$

e) Obtain the truth table

f) Implement the function using 2-input gates

g) Implement the function using 2-input gates after minimizing.

h) Compare the number of gate required in b) and c)

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7. Implement 3-bit Odd function.

8. Simplify the following function F using don't care condition d in :

$$F=w'(x'y+x'y'+xyz)+x'z'(y+w)$$

$$D=w'x(y'z+yz)+wyz$$

- Implement the function using 2-input gates

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$$\overline{w}\overline{x}+yz+\overline{w}y$$

$\frac{A}{1} + \frac{B}{1}$